THE BEE GAME

(AN OPENGL, PYGAME RACE-AGAINST-THE-CLOCK GAME)

GitHub Repo

Itch.io (Download the Game)

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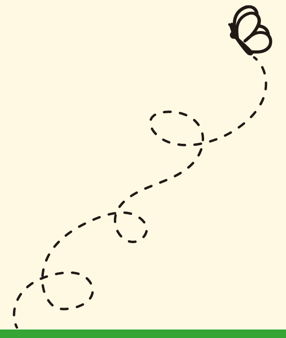
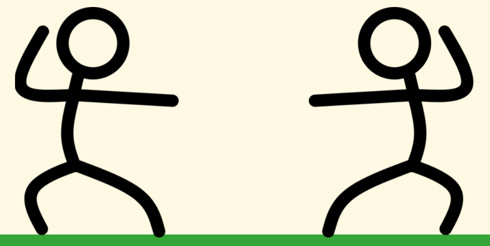
CONCEPT

You play as a friendly and hard-working bee who is ready to pollinate and explore his local garden.

1. Fly, hover, and buzz around the garden



2. Beware of enemies



3. Collect pollen particles and take them to the beehive

CONCEPT

You play as a friendly and hard-working bee who is ready to pollinate and explore his local garden!

2 "Rooms"

Lobby

- get used to the mechanics & learn how to play before starting

Level 1

- 120 seconds
- Collect pollen, attack moths (+points)
- Get caught by moths (-health)

PLAYER CONTROLS

Key	Movement	Notes
W/A/S/D	Camera Panning	• camera turn angle is bounded, depending on which view mode
Q/E	Camera Zooming	• not available in fpv or follow mode
Arrow Keys	Bee Movement	• bounded to the garden area • will face the correct direction automat.
LShift/LCtrl	Bee Ascending & Descending	• bounded to the garden area
Z	Activate Angry Mode	• lasts 5 seconds → then needs 10 seconds to recharge before re-activating
X/C	Pick Up/Drop Off Pollen	• must be colliding with the pollen particle • beehive dropoff if within range, or drops pollen to the floor

Key	Movement
P/Esc	Toggle Pause
Mouse	GUI Interaction

BEE MODES/ ANIMATIONS

You play as a friendly and hard-working bee who is ready to pollinate and explore his local garden.

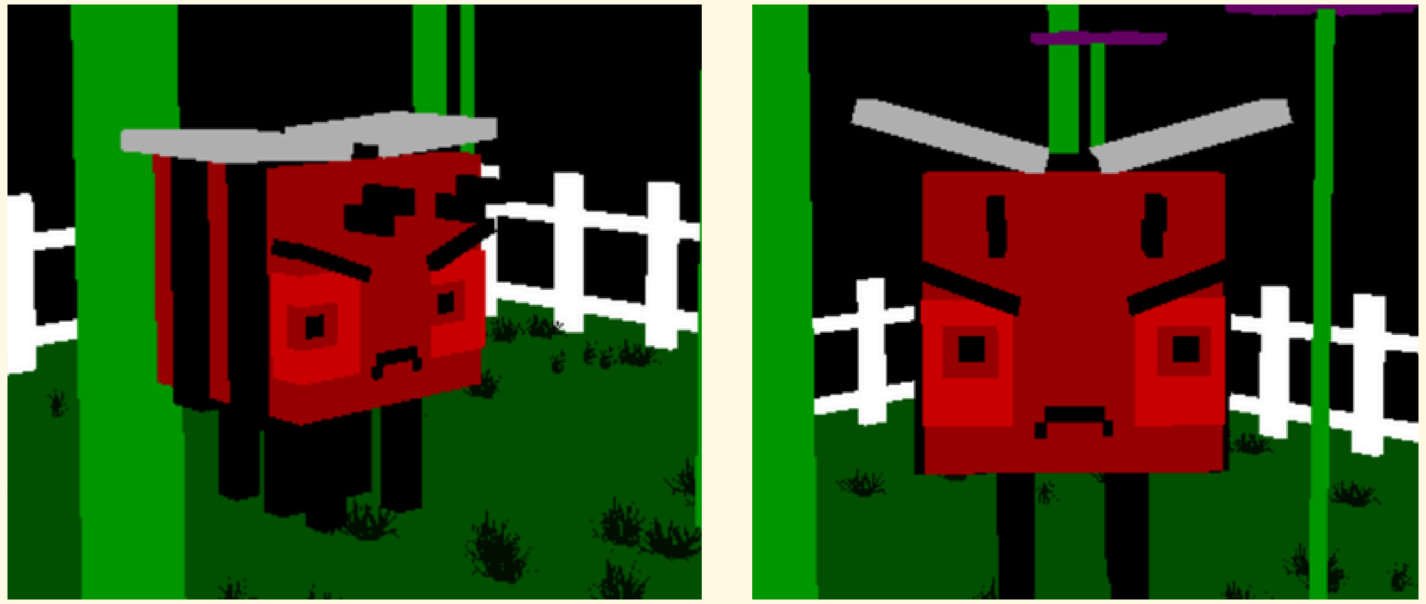
- Unique Actions 1a: Hovering
  - Legs moving in sin/cos waves to move back and forth as the bee hovers
  - Eyes moving about randomly
- Unique Actions 1b: Flying
  - Legs move in the opposite direction of the bee's movement (as if they are drifting in the wind)
- Unique Actions 2: Angry/Attack Mode
  - Colors change, eyebrows angled, pause moving eyes, faster animation and movement speed
- Unique Actions 3: Recharging
  - Only body color changes to grey
- Unique Actions 4: Carrying pollen
  - Legs angle to hold the pollen particle

BEE MODES/ ANIMATIONS

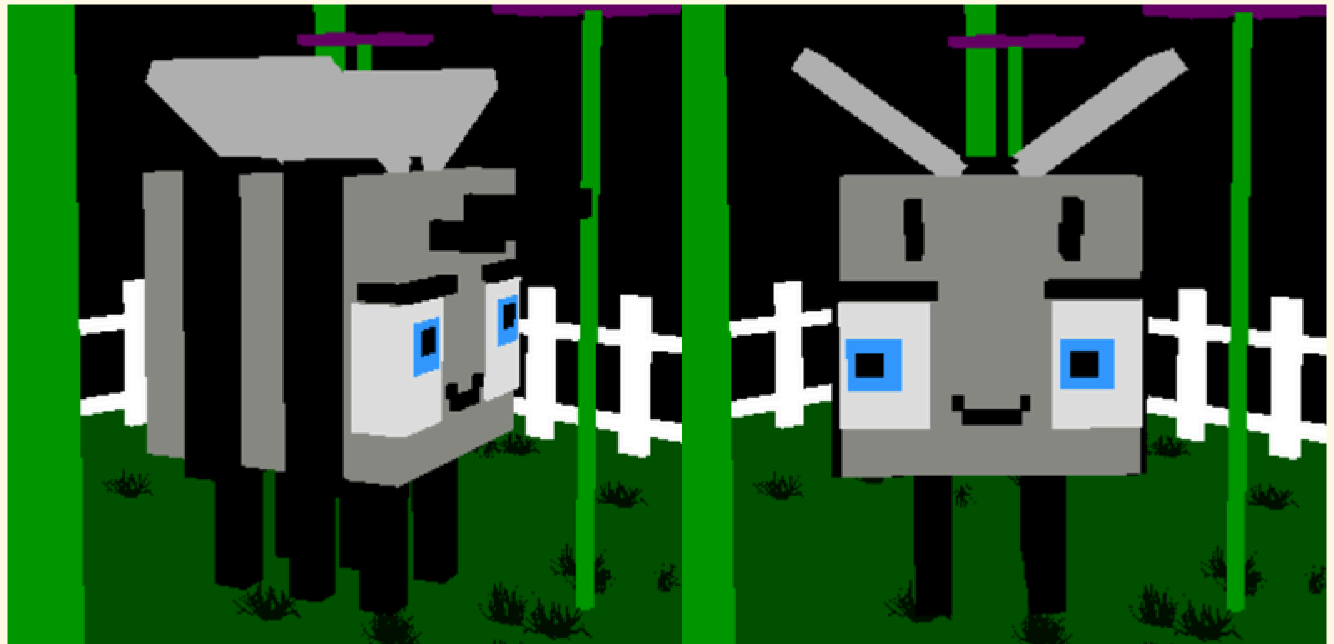
You play as a friendly and hard-working bee who is ready to pollinate and explore his local garden.

UA 1: Hovering/Flying

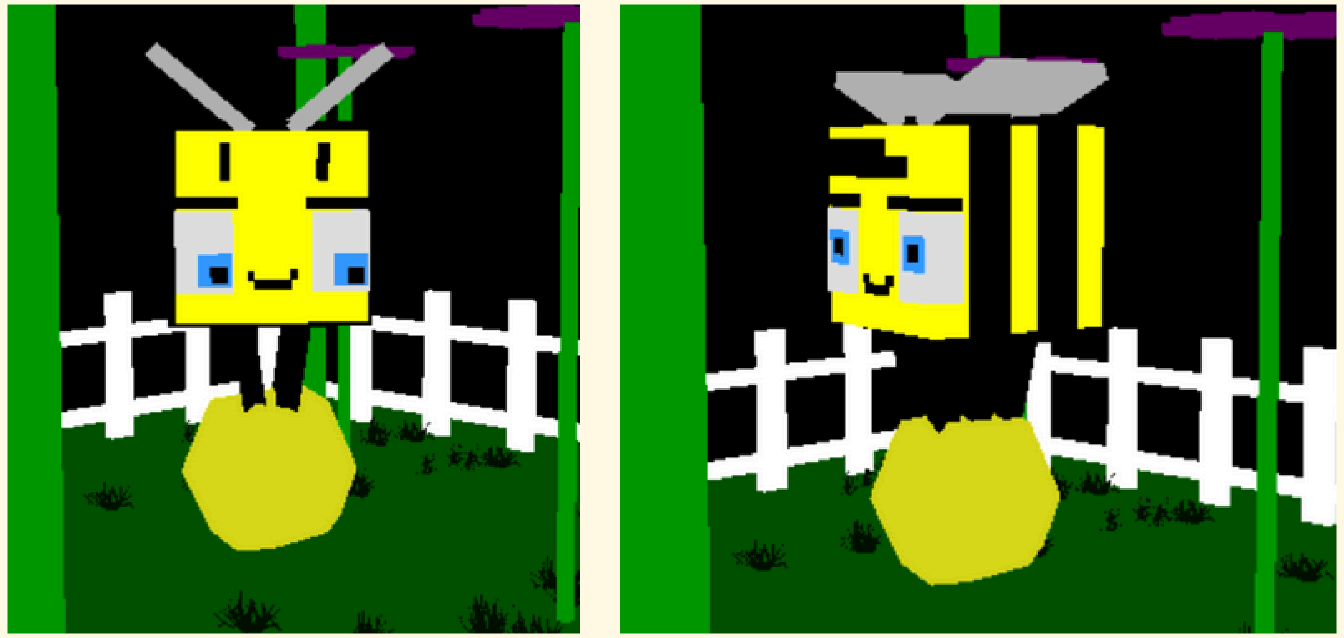
UA2: Angry/Attack Mode



UA 3: Recharging



UA 4: Carrying pollen



BEE MODES/ ANIMATIONS

COLLISIONS

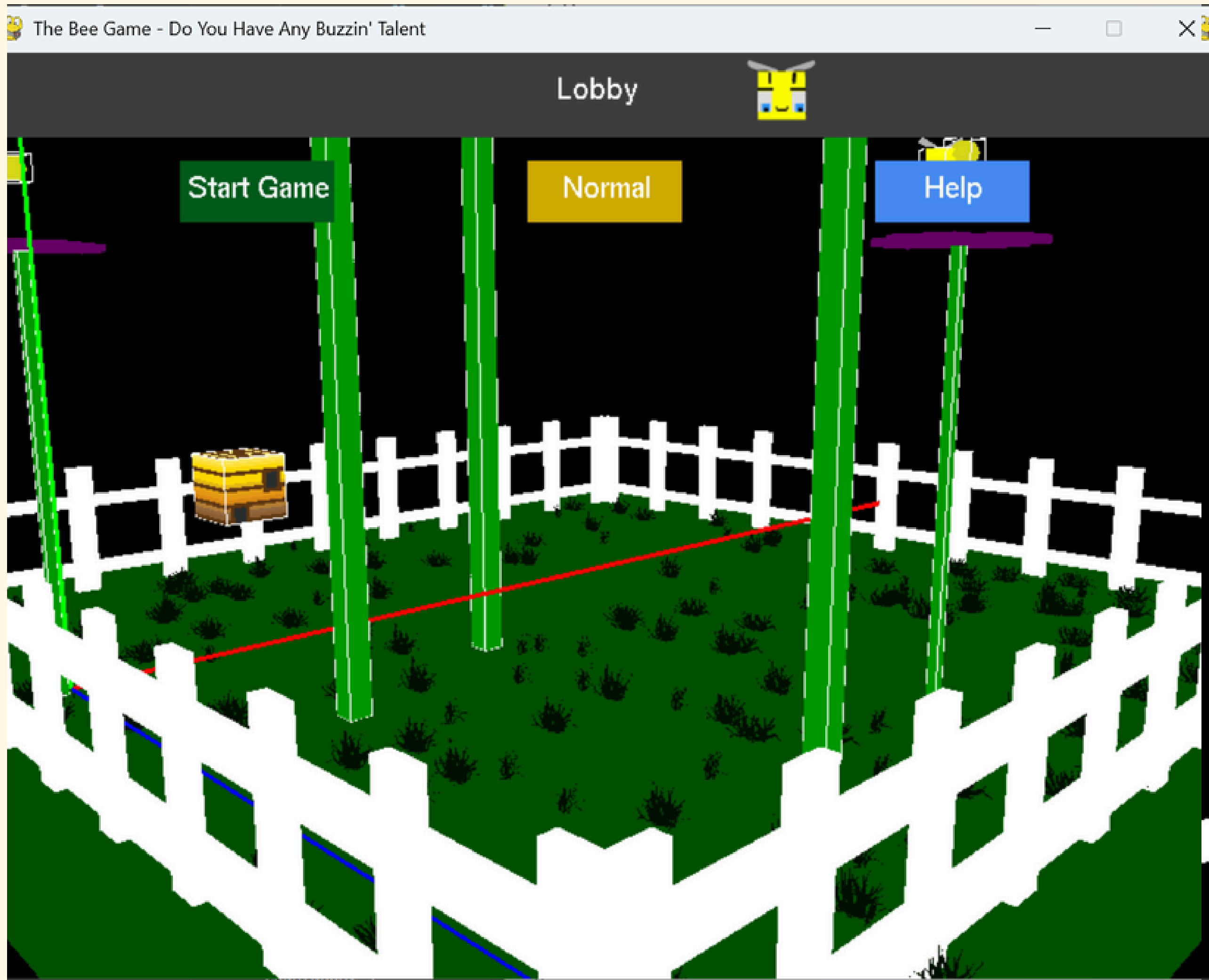
- **Bee-Props and Moth-Props**
  - Stems of the flowers and the beehive
  - Are calculated based on the current props in our list
- **Bee-Moths**
  - Moth does the calculating
  - Determines whether to decrease bee's health or increase its score based on the current state of the bee
- **Bee-Fence**
  - Simple clamping to the garden boundaries an all three axes (no AABB)
- **Bee-Pollen**
  - Bee calculates whether it is colliding with any of the pollen particles upon pressing 'x'.
  - If so, keeps track of which pollen particle it is carrying so we know which one should follow the bee and get dropped off eventually

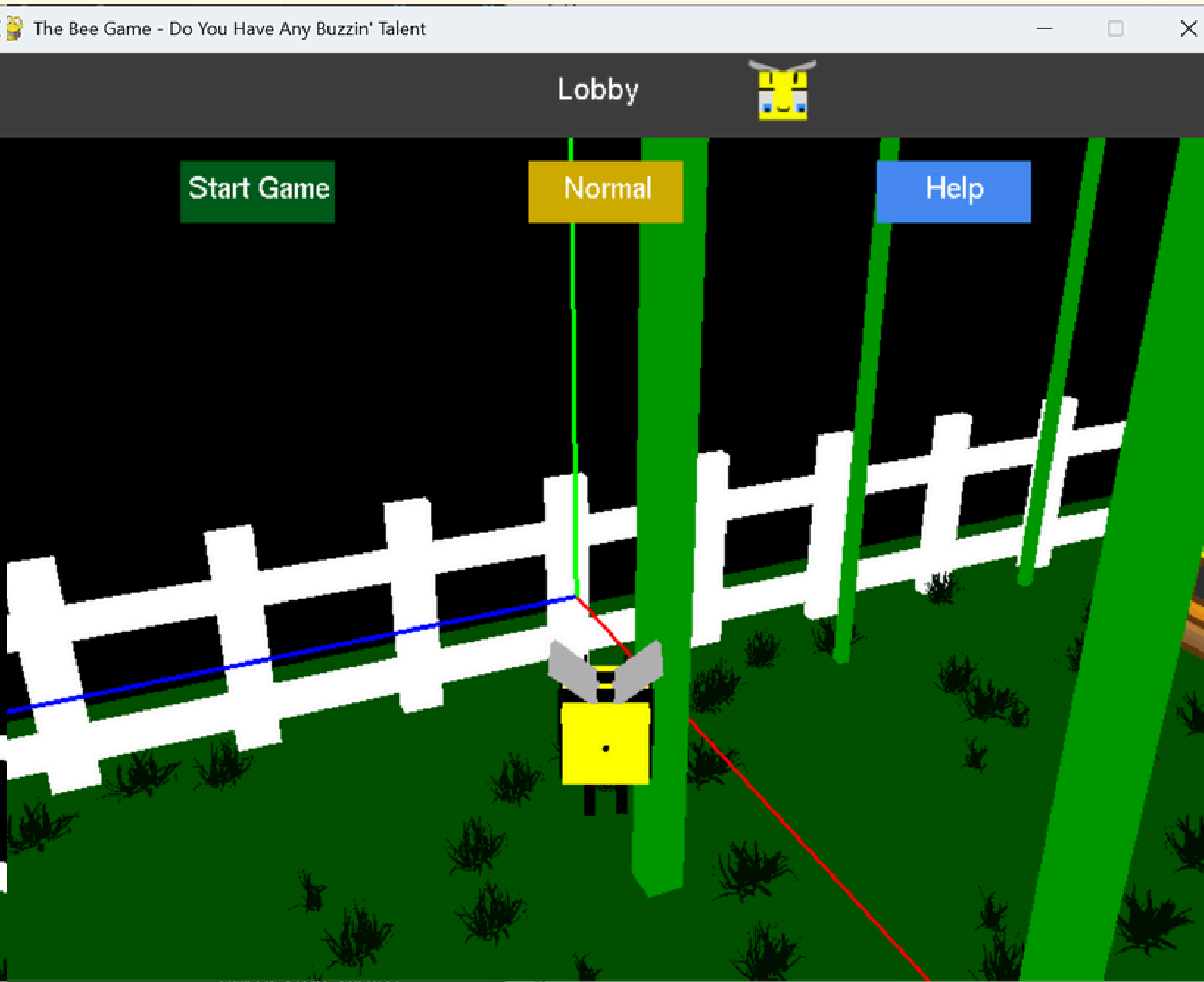
COLLISIONS


The Bee Game - Do You Have Any Buzzin' Talent
Score: 0
Health: 100%
Level:1
Sec Left: 87

CAMERA VIEWS

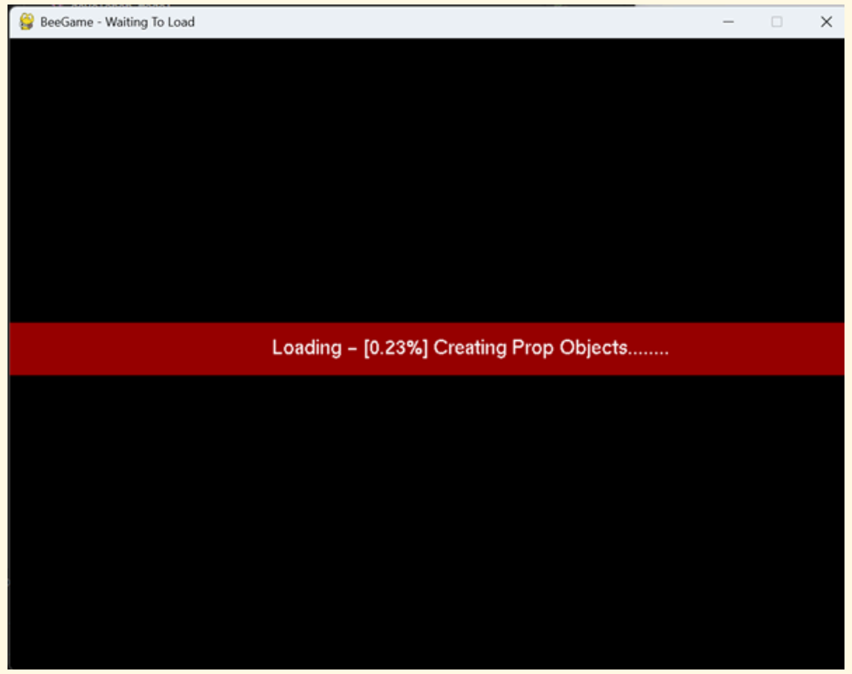
- **4 Corners**
  - at each corner of the garden and can pan around a bit, but not enough to flip over or look into the total darkness
- **FPV**
  - in the eyes of the bee
- **Follow**
  - behind and above the bee





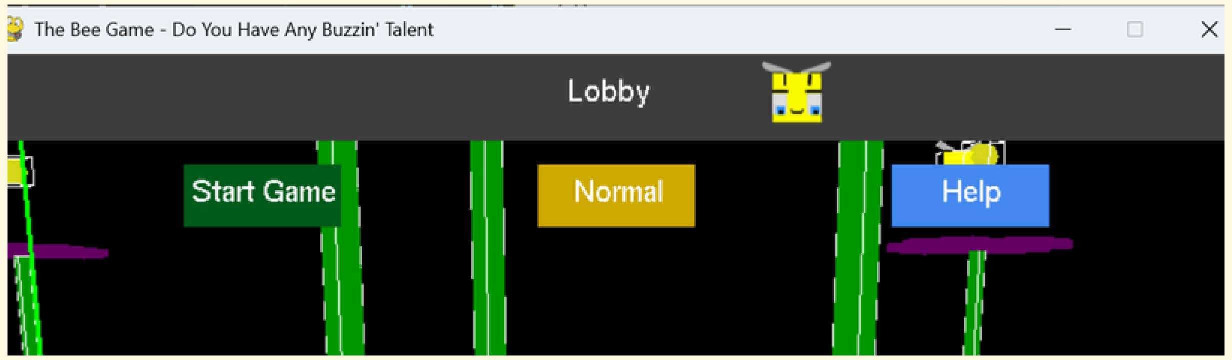
GUI

- **Loading Screens**

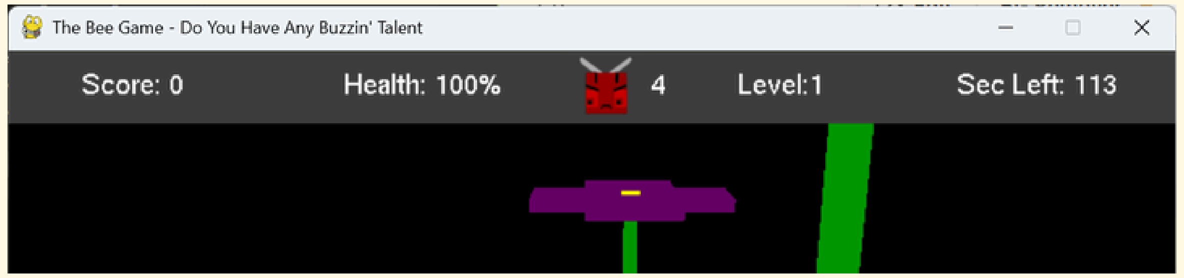


- **Uh-Oh Screen (Crash Protection)**

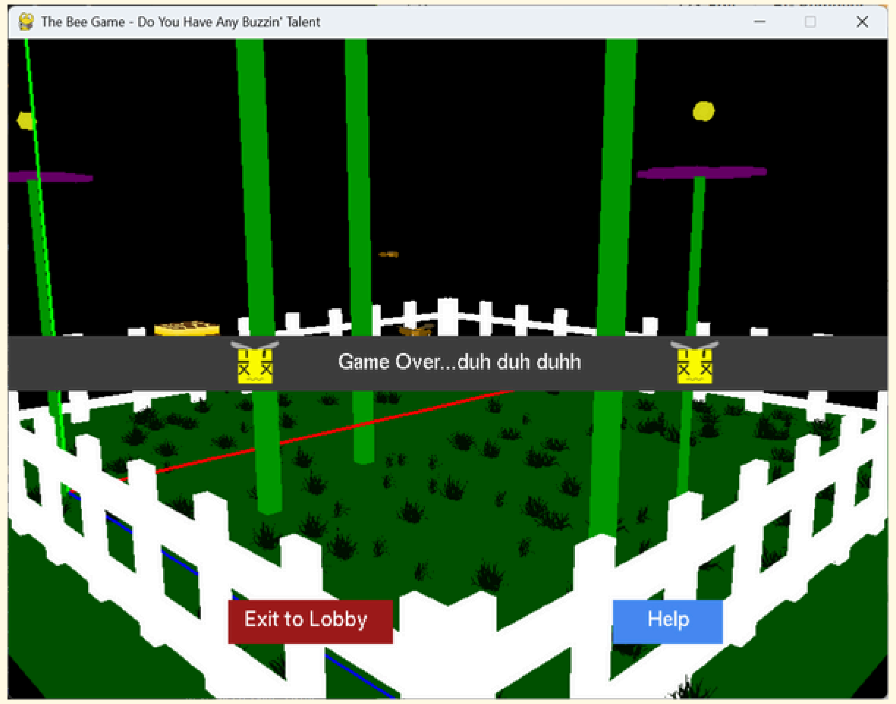
- **Lobby Screen**
  - Start Game
  - Difficulty Setting
  - Help Screen



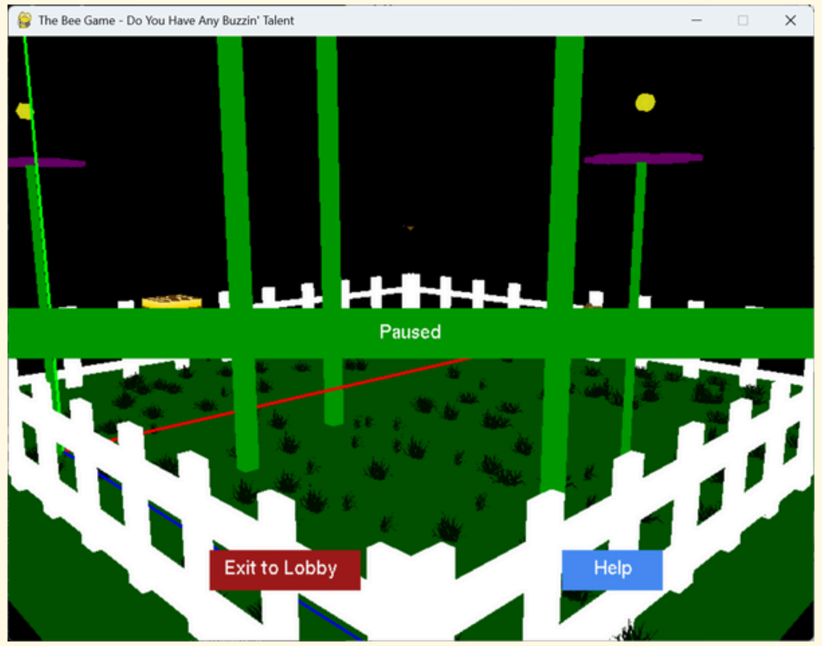
- **Level Screen**
  - Score, Health
  - Bee Mood State
  - Level, Timer

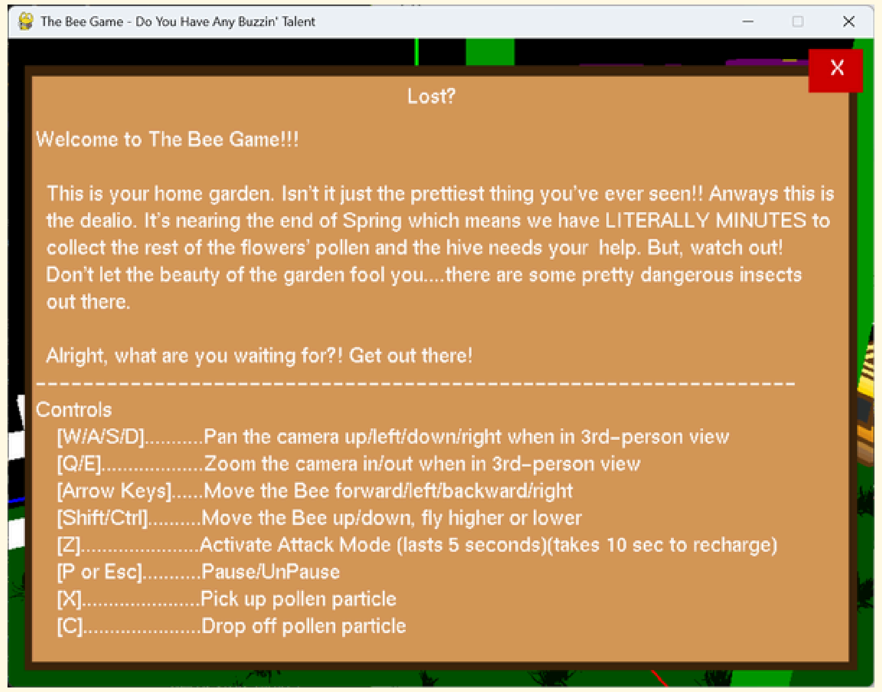


- **End of Game Screen**
  - Won/Loss Result
  - Exit to Lobby
  - Help



- **Pause Screen**
  - Exit to Lobby
  - Help





BEYOND THE SCARECROW

There is a lot of new functionality and new mechanics compared to the previous scarecrow assignment.

- Ascending/Descending
- Collision Detection
- Interactive and Responsive GUI
- Crash Protection
- Caching
- Timer
- Pause
- Character States
- Game Environment
- OBJ Props
- Loading Screen
- Lobby vs Level Distinction

CHALLENGES AND TAKEAWAYS

Biggest Challenges

- OBJ Refactoring and Debugging
- Loading Screen (needed to use threading)
- Handling the timers when in pause mode
- Collision Detection

Takeaways

- How grateful I am to have game engines like Unity and Unreal that make creating and manipulating 2d and 3d environments so easy!
- I feel now I understand how some of these tools would be implemented within the engine and can help me be a better designer and developer.
- First time I have used pygame and openGL to create complete projects and I enjoyed it.

RESOURCES

- * [Clamping Floats in Python](https://stackoverflow.com/questions/9775731/clamping-floating-numbers-in-python)
- * [Readme Formatting](https://docs.github.com/en/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax#links)
- * [RGB Values - Shades of Blue](https://www.rapidtables.com/web/color/blue-color.html)
- * [Minecraft Bee Inspo](https://minecraft.wiki/w/Bee)
- * [Type Casting for Function Parameters](https://stackoverflow.com/questions/2489669/how-do-python-functions-handle-the-types-of-parameters-that-you-pass-in)
- * [GLUT Bitmap Fonts](https://www.opengl.org/resources/libraries/glut/spec3/node76.html)
- * [GLUT Solid Sphere Function](https://www.opengl.org/resources/libraries/glut/spec3/node81.html)
- * [3d Graphing Tool to Visualize Paths](https://www.desmos.com/3d)
- * [3d Visualization of All Moth Paths](https://www.desmos.com/3d/fb2pxzfkns)
- * [RGB 0-1 Color Picker](https://rgbcolorpicker.com/0-1)
- * [Photoshop Express Adobe Desktop Version](https://new.express.adobe.com)
- * [Rendering Images in PyGame and OpenGL] (https://www.reddit.com/r/pygame/comments/17ob2qc/texturing_in_pyopengl_and_pygame/)
- my handy notebook and whiteboard at home where I drew so many grids to get the right transformations

LIVE DEMO

LEFT TO DO

(as of 5/16/25)

- README
- Clean up this presentation
- Clean up the code and review comments
- Clean up my github repo
- Credits Screen in the Lobby
- Demo Video

ADDITIONAL IMPLEMENTATION DETAILS

(More detailed info on certain mechanics)

POTENTIAL ADDITIONS

- **Deployed to Personal Website**
- **More Differentiation Between Difficulties**
- **More than 1 Level**
    - 2 Routes:
        - [1] - Save user progress, level select screen reflects what levels you have passed (would involve hard coding differences between levels)
        - [2] - Endless, keep advancing until you lose all your health. Personal best saved instead of level select. (would involve more automatic generation)
- **More Items in the Garden**
    - Harmless Garden Decor
    - New Types of Enemies, Flowers (thorns), Interactions (ex: wind tunnel gives you a boost mario-kart-style)
- **Attack Animations for Bee and Moths/Other Enemies**
- **Two Player Possibilities?**
- **Lighting** (although I don't think the game needs it)
- **Sounds?**

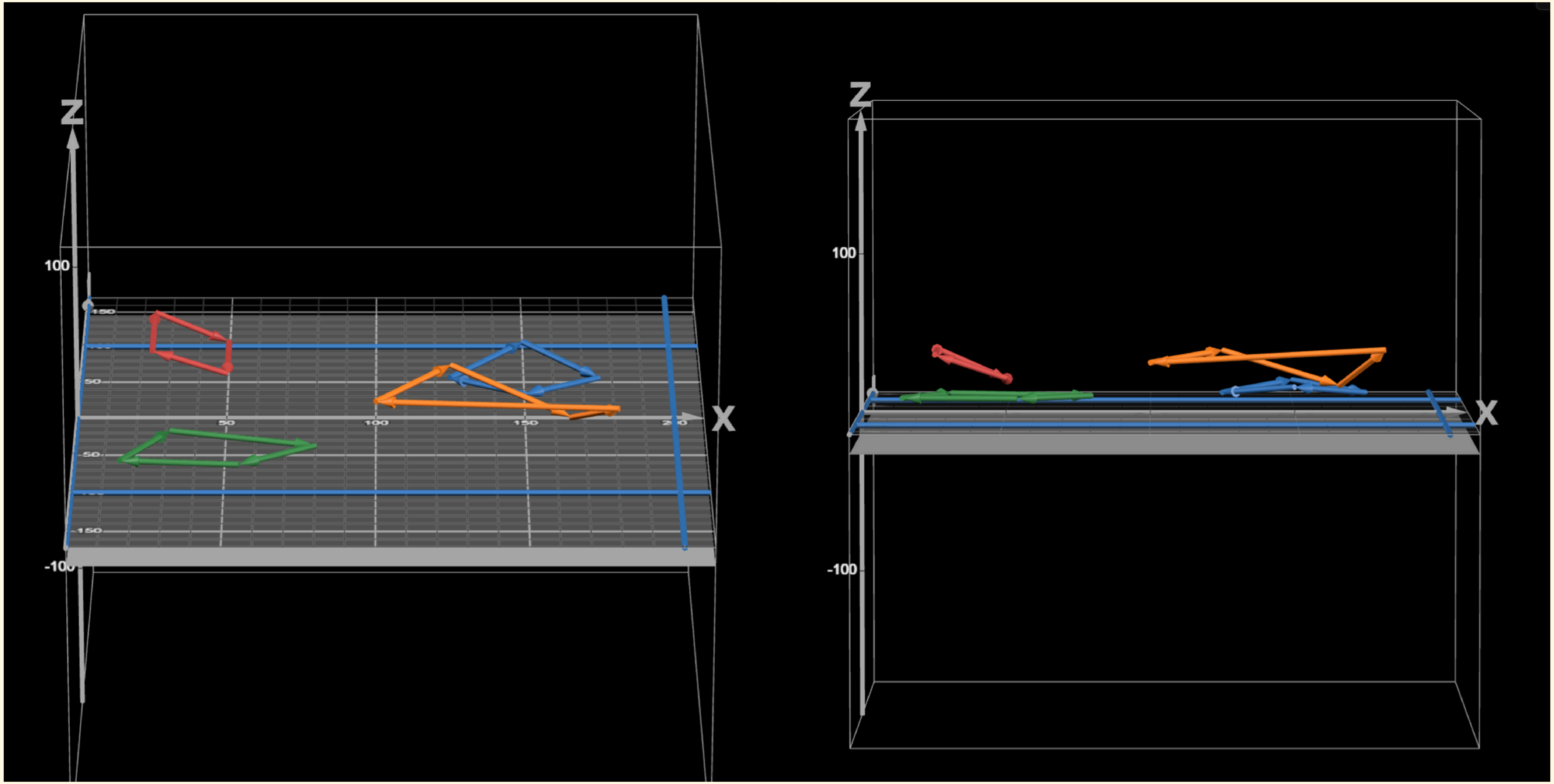
DEVELOPER MODE

Toggle With DELETE Key

- 1 – Decrease Animation Speed
- 2 – Increase Animation Speed
- 3 – Decrease Fly Speed
- 4 – Increase Fly Speed
- 5 – Decrease Health by 20
- 6 – Increase Score by 50
- 7 – Toggle Show Colliders (AABBs)
- 8 – Regenerate flowers
- 9 – Crash the Game (throws an error)
- u – Place Bee in front of Corner Camera 1

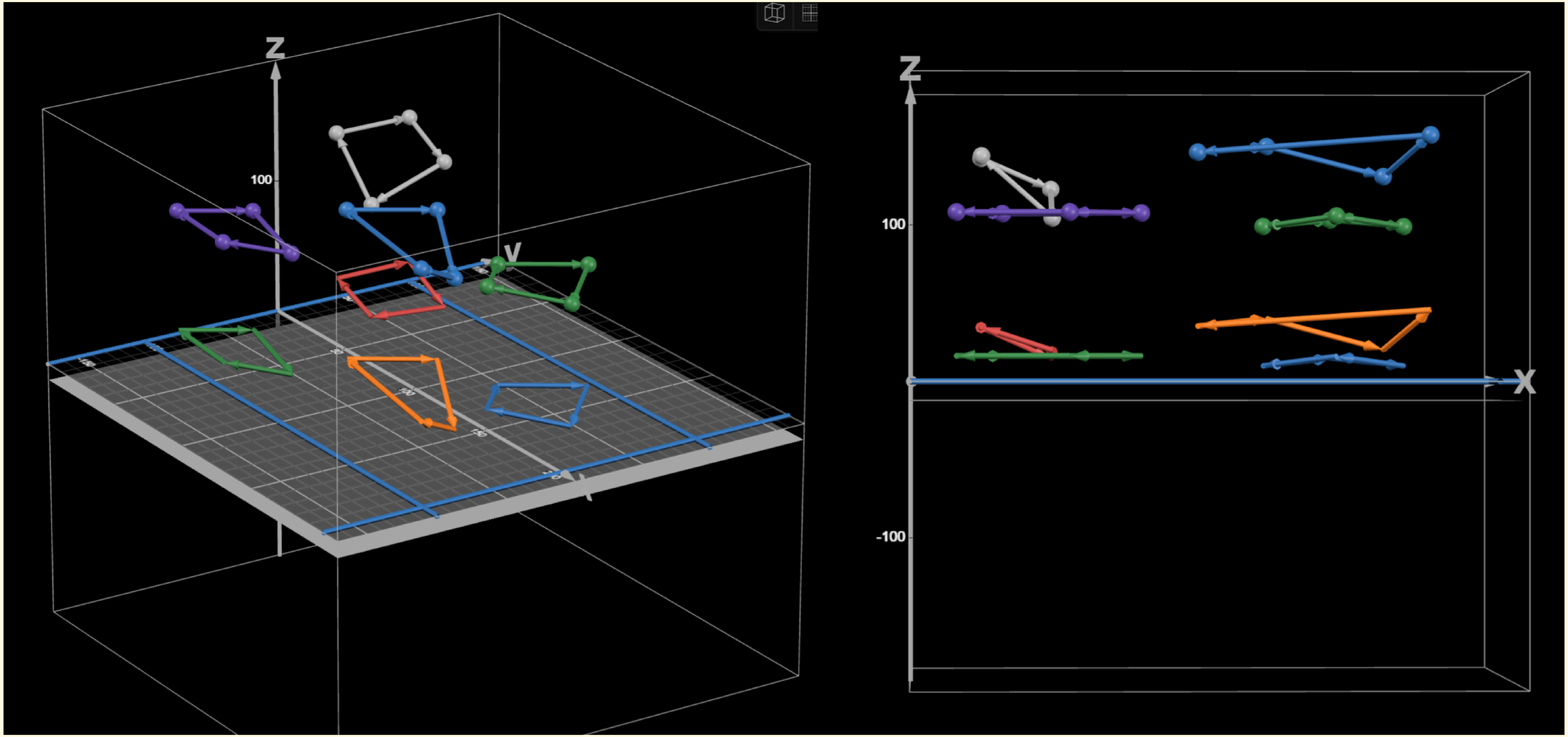
DIFFICULTIES - NORM

desmos moth paths visualized



DIFFICULTIES - HARD

desmos moth paths visualized



MOTH BEHAVIOR

Following Patrol Route

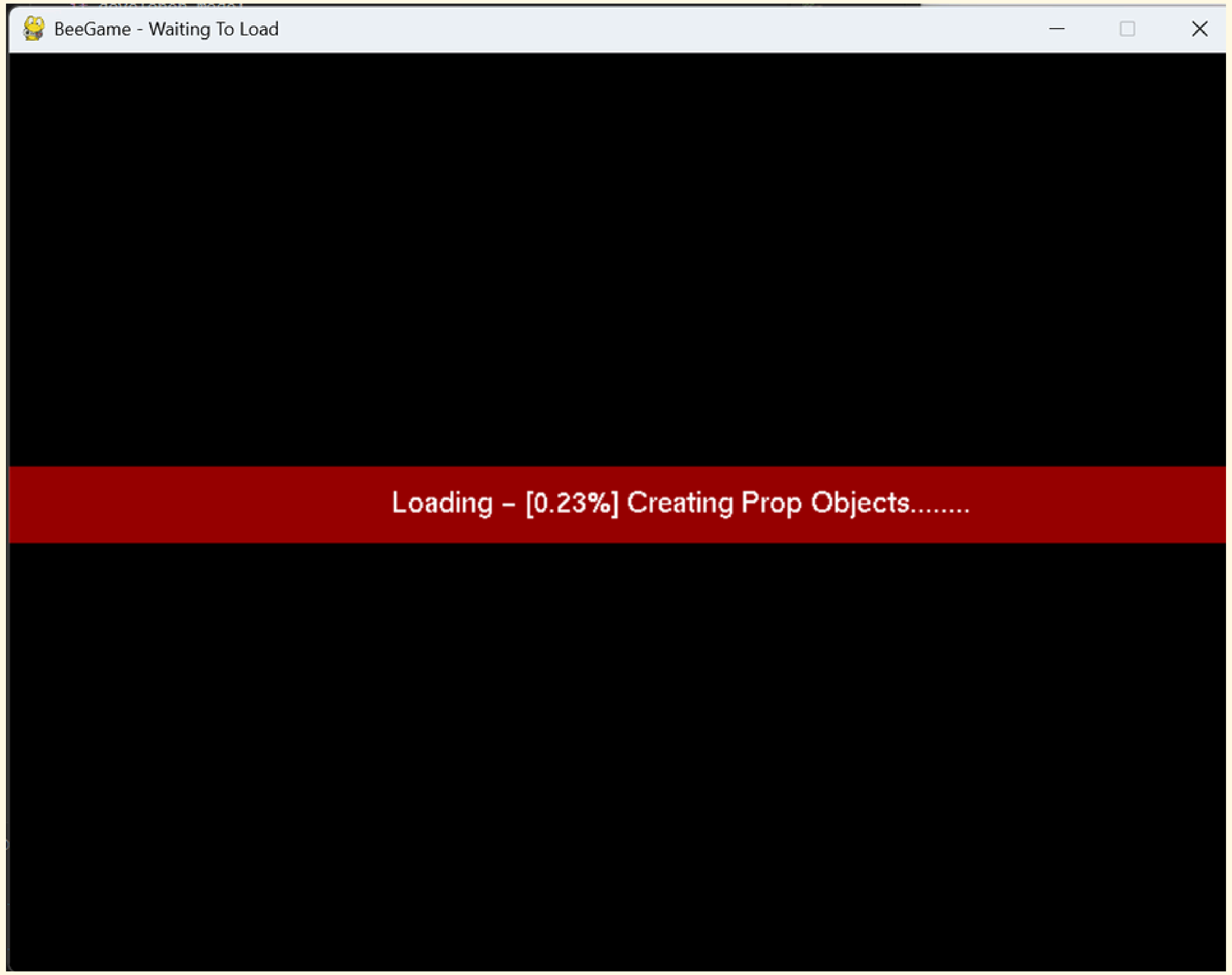
- Always have a target position
- Make their way to that target position until they detect collision with it
- Always detecting collision with props (distance of 1 from their center)

Chasing Bee Mode

- The Bee's position is their target
- If within 3 of the bee, they have collided with them
    - if bee is angry, moth goes back to spawn point to patrol, bee earns points
    - if bee is not angry, moth goes back to spawn point, bee loses health

LOADING SCREEN

- A thread is set up to do all the file I/O for the grass positions
  - While this thread is running, the first loading screen will appear
- Then, as it loops through the grass positions and instantiates Prop objects, the second loading screen will reload with the current percentage of progress
- Events are "pumped" to the game during both of these loops to prevent the OS from thinking the application is frozen.
  - This also means that you can press Esc at any point to quit the program.
- Final loading screen runs for 2 seconds after the second loop is done and it used for aesthetics and not because it is masking anything else happening.



OBJ ISSUES

While not necessarily an issue, loading in obj files quickly slowed down the program and caused everything within the game to move much slower.

I made the decision to instead model my own flowers, but keep the obj imported grass models.

The obj loading functionality is complete, but caused too far an issue in terms of efficiency. Therefore, additional models should be made from OpenGL primitives rather than imported highly complicated obj files.

QUESTIONS?